

Natural Gas Market Structure & Basis Trading Framework
AECO vs Henry Hub | Regime-Based Supply & Trading Model

Economist: Marcus Williams
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Takeaway

Storage-driven regime shifts explain a significant portion of AECO–Henry Hub basis volatility and provide a repeatable framework for directional bias in short-term natural gas positioning.

The model maps storage stress and seasonality directly into regime-based signals that capture periods of pricing dislocation and regional imbalance.

1. Objective

Framework maps storage, seasonality, and volatility into regime-based directional bias for Henry Hub and AECO basis trading.

It is designed to:

- Identify storage-driven market stress regimes
- Capture directional bias in HH pricing
- Detect AECO–HH basis dislocations
- Flag volatility expansion during physical imbalance

The output is a simplified representation of how physical market stress translates into tradable price signals.

2. Market Structure Overview

Natural gas pricing is driven by recurring supply-demand imbalances and physical constraints across North America:

- Regional supply-demand dislocation (AECO vs Henry Hub)
- Storage tightness / surplus cycles
- Seasonal demand shocks (winter heating / summer injections)
- Weather-driven consumption volatility (HDD/CDD swings)
- Pipeline and transport constraints (regional bottlenecks)
- Power-sector gas switching dynamics
- Liquidity and short-term positioning effects

These drivers collectively create periodic pricing dislocations, volatility clustering, and basis instability across hubs.

3. Core Analytical Constructs

Basis Spread (AECO – Henry Hub)

Measures regional pricing dislocation between Canadian and U.S. hubs.

- Widens during supply constraint or regional imbalance
 - Compresses during balanced or well-supplied conditions
 - Functions as a direct proxy for physical flow friction and regional stress
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Storage Regime

Primary indicator of physical market tightness.

- High storage → oversupply → bearish pricing pressure
 - Low storage → tightening system → bullish repricing
 - Rate-of-change signals stress more accurately than level
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Seasonality (Structural Imbalance Driver)

- Winter: structurally tight system → bullish bias
 - Summer: injection-driven → softer pricing pressure
 - Transitions: volatility expansion + regime instability
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Volatility Regime

Captures market stress and repricing phases:

- High volatility → regime transition / supply shock / imbalance
- Low volatility → equilibrium / stable fundamentals

4. Key Insight

Pricing dislocations are most pronounced when storage stress aligns with seasonal transitions, creating sharp repricing in both AECO and Henry Hub.

These conditions reflect:

- Physical tightness or oversupply
- Regional constraint amplification
- Fast-moving expectation shifts

This is where basis and directional gas trades become most effective.

5. Trading Implications

The model translates directly into positioning logic:

- Bullish regime (low storage): Long Henry Hub exposure / tightening bias in regional balance
- Bearish regime (high storage): Short Henry Hub / oversupply-driven pressure
- Basis widening environment: AECO underperformance vs Henry Hub (regional stress trade)
- Volatility expansion regimes: Reduce size / shift to short-dated or tactical positioning

Positions are adjusted for volatility to reduce exposure during unstable regimes.

6. Model Framework

Inputs:

- Henry Hub prices
- AECO proxy prices
- Storage levels

Feature Set:

- Normalized price series
- Rolling storage Z-score
- Volatility regime indicator
- Seasonal classification

Outputs:

- Regime signal (Bullish / Bearish / Neutral)
- Basis dislocation indicator
- Directional bias signal
- Backtested performance

7. Results & Interpretation

Figure 1 — Strategy vs Benchmark (Cumulative PnL)

Strategy outperformance is driven by correctly identifying storage-driven regime shifts rather than directional price prediction. Returns cluster during periods of physical market stress and seasonal transition.



Figure 2 — AECO vs Henry Hub Spread (Basis Dynamics)

Spread widening reflects regional supply-demand imbalance and transport constraints. Periods of divergence align with structural stress in AECO relative to Henry Hub, indicating localized oversupply or constrained takeaway capacity.

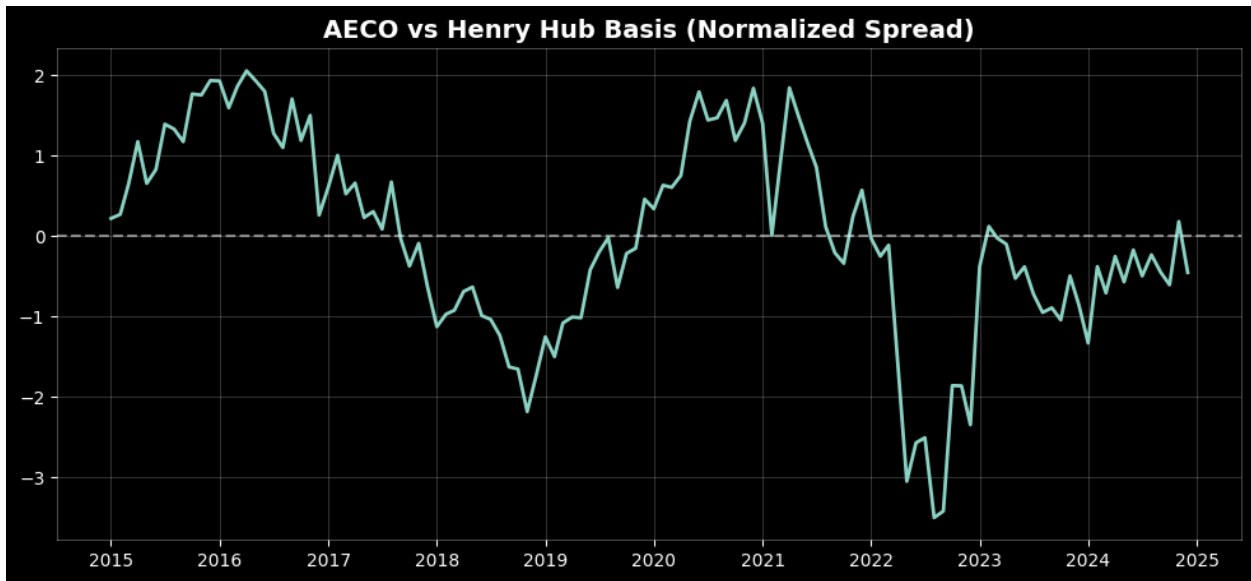
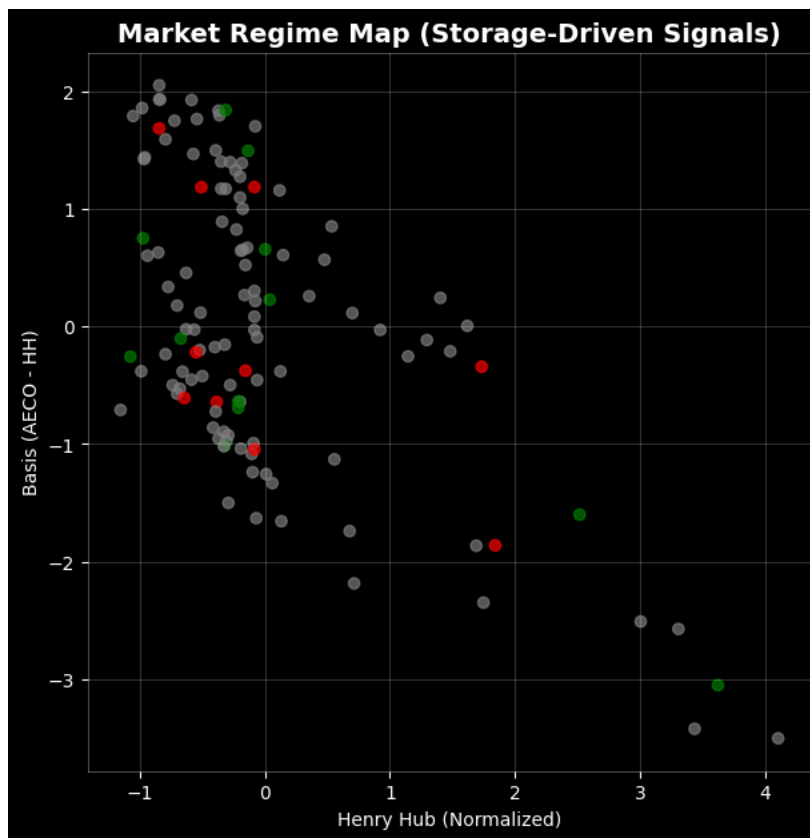


Figure 3 — Regime Map (Market State Classification)

Regime clustering shows clear separation between bullish and bearish storage conditions. Transition zones correspond to elevated volatility and unstable pricing environments, where directional signals are most active.



8. Application to Supply & Trading

This framework is directly applicable to:

- Gas procurement optimization
- AECO basis risk management
- Generation fuel cost hedging
- Short-term directional trading
- Fundamental overlay for discretionary trading decisions

9. Conclusion

This model provides a structured mapping between physical gas fundamentals and tradable market behavior.

Rather than forecasting price directly, it focuses on:

- regime detection
- storage-driven stress identification
- seasonal imbalance timing
- basis dislocation capture

The result is a repeatable, interpretable trading framework aligned with real supply and trading desk logic.